4 Software

1

Software	System (✓)	Application (✓)
Photo editing software		✓
Graphics manipulation software		✓
Compiler	✓	
Spreadsheet software		✓
Printer driver	✓	
QR code reader		✓
Anti-virus software	✓	
Screensaver	✓	

2 **a** Three from:

- It is a set of programs to control and manage the operation of computer hardware.
- It provides a platform on which software can run.
- It is required to allow hardware and software to run without problems.
- It provides a human computer interface (HCI).
- It provides the control of allocation and usage of hardware resources.

b Three from:

- It is used to perform various applications.
- It allows a user to perform specific tasks.
- It may be a single program or a suite of integrated programs.
- The user can execute the software whenever needed.

c Three from:

- virus checkers
- defragmentation software
- disk contents analysis/repair
- file compression
- file management
- back-up facility
- security of the system
- screensaver

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3
- antivirus
- background
- anti-virus
- heuristic checking
- quarantine
- user

- blocks
- sectors
- tracks
- disk defragmenter
- blocks
- contiguous
- head movements

- screensavers
- security systems
- screensaver
- locked

4 **a** Five from:
- human-computer interface
- multitasking
- error handling
- platform for the running of applications software
- management of user accounts
- managing files
- hardware management
- memory management
- interrupt handling
- security management

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b i CLI:

- command line interface
- user uses a keyboard to type in commands/instructions …
- … to choose from a menu, open software, and so on
- requires a number of exact commands to do a single task
- user has to learn the commands …
- … and type them in accurately

GUI:

- graphical user interface
- user interacts with the computer using onscreen icons rather than typing commands
- chooses from menus, open software, and so on, by clicking on an icon (using pointing device) or by using finger on a touch screen
- no typing needed
- often uses WIMP environment
- where a windows manager looks after interaction between windows and applications software

ii

Statement	**CLI (✓)**	**GUI (✓)**
The user is in direct communication with the computer	✓	
The user is limited to the icons shown on the screen		✓
The user needs to learn a number of commands to carry out any operation	✓	
Commands need to be typed in using the correct format and spelling	✓	
There is no need for the user to learn any commands to use the interface		✓
The interface needs a complex operating system, such as *Windows*, to operate, which uses considerable amounts of memory		✓

5 a

- It allows computers to carry out many tasks/processes at a time.
- Each process will share hardware resources under the control of the OS.
- It monitors:
  - allocation of resources to a process for a specific time limit
  - interruption of processes while running
  - prioritisation of a process for resource allocation

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b
- allows more than one user to log onto the system
- allocates users a part of memory protected by passwords and user IDs
- allows a user to customise

c
- ensures integrity, confidentiality and availability of data
- carries out operating system and applications software updates as required
- ensures antivirus (and other security software) is always up to date
- communicates with the firewall to check incoming and outgoing traffic
- maintains privileges to prevent users from entering 'private areas'
- maintains user access rights
- allows for data recovery following loss or corruption of data
- helps prevent illegal intrusion into a computer system

6 a i It is a signal sent from a device or software to a CPU/processor requesting attention. The CPU/processor suspends all operations until the interrupt has been serviced.

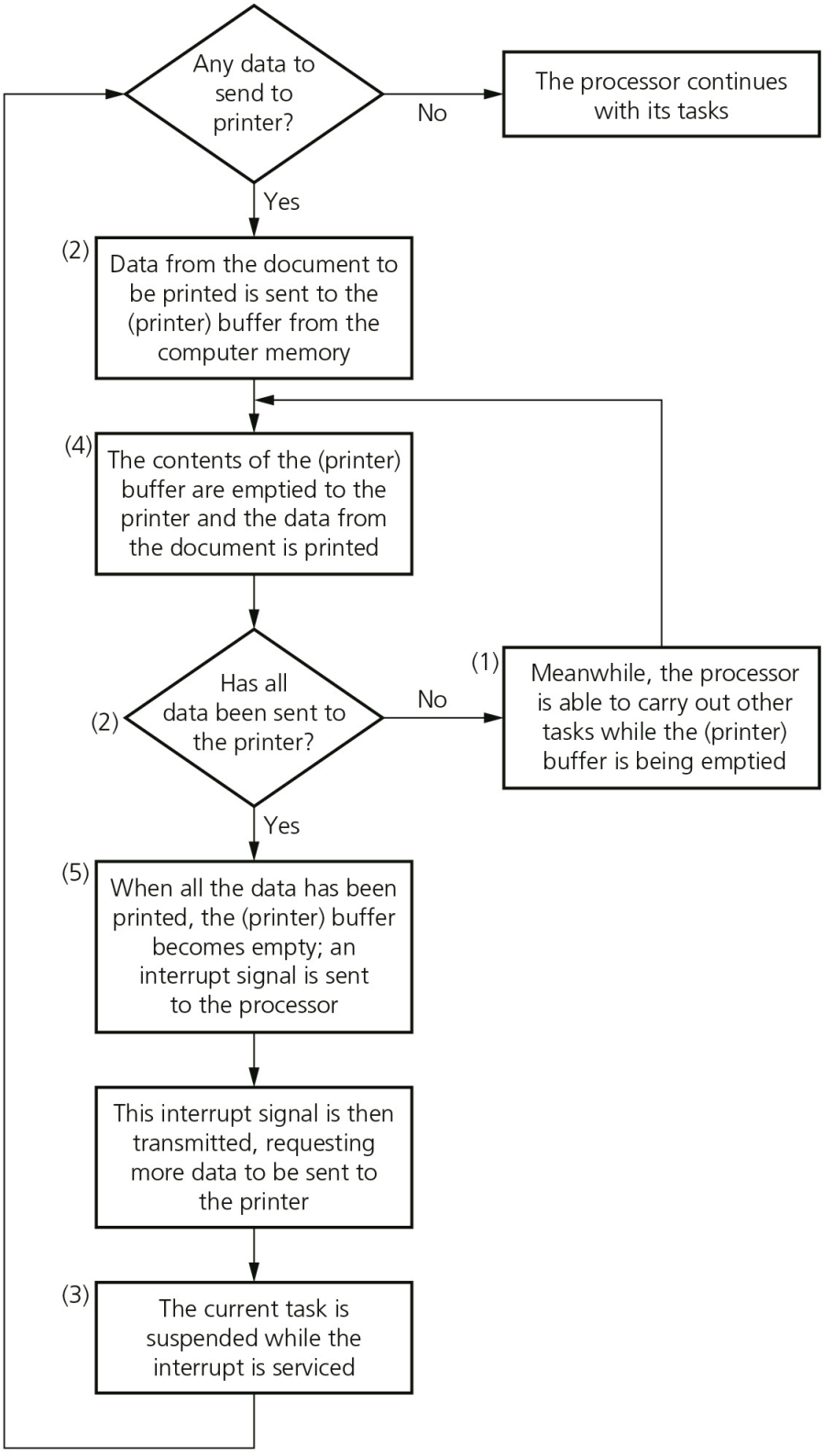
ii Three from:
- a timing signal
- I/O process (for example, a disk drive or printer requesting more data
- a hardware fault (for example, a paper jam in a printer)
- software errors (for example, can't find .exe file or attempt to divide by zero)
- operator input (for example pressing <BREAK> key

b i a memory area which stores data temporarily

ii Devices run at a slower speed than the CPU/processor:
- Buffers allow devices to carry out their functions while the CPU can carry on with other tasks, therefore greatly improving efficiency, because the CPU is no longer left idle waiting for an output/input device waiting to complete a task.
- Buffers are needed, for example, when streaming from the internet because the internet download speed may be slower than a device's playback speed and a buffer gives temporary storage which fills up allowing data to be available for smooth uninterrupted playback.

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c



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7

	Management type:		
Statement	**Memory**	**Security**	**Hardware**
Keeps a track of all memory locations	✓		
Ensures that anti-virus software is regularly updated		✓	
Prevents competing applications using the same memory locations at the same time	✓		
Ensures that the appropriate device driver takes data from memory/file and translates it into a format the device can understand			✓
Management of devices to control the use of queues and buffers holding data temporarily			✓
Receives and handles error messages and interrupts from devices connected to the computer			✓
Manages RAM and allows data to be moved between RAM and devices, such as HDD or SSD	✓		
Maintains access rights for all users of a computer system		✓	

8 **a** It is software that communicates with the OS and translates data into a format understood by the device.

b and **c**

- a collection of information about devices; known as descriptors
- allows the USB bus to ask a newly connected device what it is
- descriptors include vendor ID, product ID and unique serial number

9 **a** software/instructions programmed on a hardware device

b BIOS, videocard, SSD, CD/DVD drive are common examples.

10 **a** In any order:

- Programs are easier to write.
- Programs take less time to write.
- Programs are easier to maintain.

b A programmer may be writing a program that needs to perform a task very quickly. If he uses a low-level language, he can ensure only essential programming statements are included. This would reduce the time taken to execute the task.

c Type of language: assembly language

Reasons: Steve is writing a program that uses specialist hardware to control the lighting and there are no high-level programming commands available to drive this hardware.

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11
- language
- machine code
- assembler
- high-level
- language
- machine code
- translators
- high-level
- compilers
- interpreters

The last two answers can appear in either order.

12 a A compiler translates a program written in a high-level language into a machine code program. The whole program is translated.

b An interpreter executes a program written in a high-level language by translating, then executing each line of the program.

c An assembler translates a program written in assembly language into a machine code program. The whole program is translated.

13 a i to translate the final program that is going to be used repeatedly

ii while the program is being developed

b In any order:
- It is easier to edit programs during development.
- It is faster to debug programs during development.

c In any order:
- Once translated, programs do not need the compiler to execute and only need translating once.
- Once translated, programs take up less space in memory during execution.

14 a An integrated development environment (IDE) is used by programmers to aid in the writing, editing and development of programs.

b Three from:
- code editor
- translator
- auto-completion
- auto-correction
- pretty printing

c Error diagnostics and debuggers allow the programmer to run the program under development in a controlled environment. For example, a separate report window can be used to display the values of each variable as the program is running.

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